

Artificial Intelligence 2: Bayesian Networks I – Introduction and Representation

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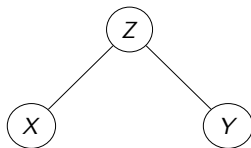
Outline of Topics

- 1 Introduction to Bayesian Networks
- 2 Bayesian Networks: Representation
- 3 BN structures as probabilistic relationships
 - Conditional Independence

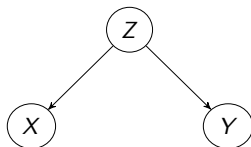
Probabilistic Graphical Models

Probabilistic graphical model: graphs which nodes represent random variables, and the edges (aka. links or arcs) represent **conditional independence** assumptions (explained later). According to the edges, we have:

- **Undirected graphical models:** Markov Random Fields (MRFs, or aka Markov networks)



- **Directed graphical models:** Bayesian Networks ✓



Bayesian Networks

Bayesian Networks: a kind of probabilistic graphic models that uses the direction of edges to represent the cause-effect relationship and Bayes' theorem for probabilistic inference.

Real-world applications of Bayesian Networks:

- Legal Tech: Bayesian networks for settling legal disputes, see [ArbiLex, A Harvard Law School Legal Tech Startup, Uses AI To Settle Arbitrations](#)
- Chemistry: prediction of optimal reaction rate and energy, see [Explainable and trustworthy artificial intelligence for correctable modeling in chemical sciences](#)
- Cyber security – [Darktrace](#), a British cyber security AI company that reach a valuation of \$1.65 billion. [Everything you need to know about Darktrace and their patent](#)

Bayesian Networks: pros and cons

Advantages:

- Graphical representation: Provide a visual representation of joint probability distributions of different random variables – **interpretability**
- Powerful: can capture complex relationships between random variables
- Combine data and prior knowledge: Prior knowledge can be incorporated and updated with statistically significant information from data – better approximation of true knowledge.
- **Generative approach**: generate new data similar to existing data

Disadvantages:

- Require prior knowledge of many probabilities.
- Sometimes computationally intractable.

Example 1: Naïve Bayes as Bayesian Networks

The simplest Bayesian network – Naïve Bayes Classifier

- Dependent variable Y (Class label)
- Independent variables $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ (Evidence)
- Main assumption: independent variables are conditional independent given $Y = y$

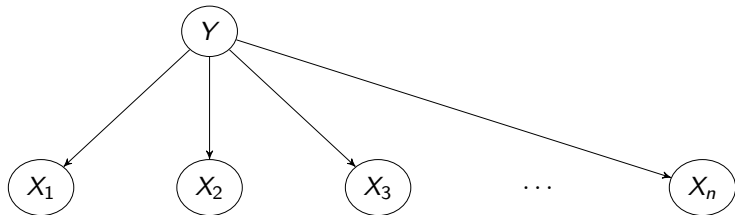


Figure 1: Graphical representation of a Naive Bayes Classifier

Main problems in Bayesian Networks

Wet grass

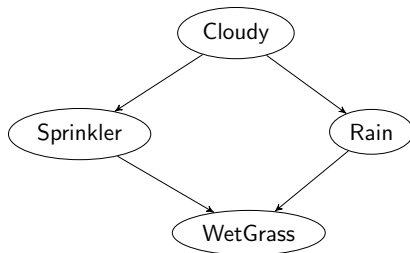


Figure 2: A famous example of Bayesian Networks

Three main problems with Bayesian Networks:

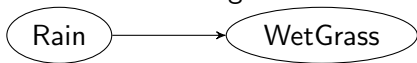
- Inference (from observations “it’s cloudy” infer the probability of the wet grass) ✓
- Training the models ✓
- Determining the structure of the network (i.e. what is connected to what)

Bayesian Networks: Representation

Problem: How to represent the joint probability distributions of random variables.

Solution: A Bayesian network – a directed, acyclic graph, which consists of

- a set of nodes: each represent a random variable
- a set of directed edges that connect those nodes, for example:



, the directed edge represents “directed dependency” or “directed influences”, also called “direct cause”

- A conditional distribution for each node given its parents:

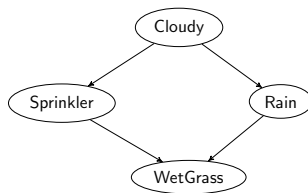
$$P(X_i | \text{Parents}(X_i))$$

- Discrete Random Variables: conditional distribution can be represented as a conditional probability table (CPT) – the distribution over X_i for each combination of parent values

Bayesian Networks: Wet grass example

$P(C = 0)$	$P(C = 1)$
0.4	0.6

C	$P(S = 0)$	$P(S = 1)$
0	0.5	0.5
1	0.9	0.1



C	$P(R = 0)$	$P(R = 1)$
0	0.8	0.2
1	0.2	0.8

S	R	$P(W = 0)$	$P(W = 1)$
0	0	1.0	0.0
1	0	0.1	0.9
0	1	0.1	0.9
1	1	0.01	0.99

Bayesian networks: the equation

Full joint distribution:

$$\begin{aligned}
 P(X_1, X_2, \dots, X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots P(X_n|X_1, X_2, \dots, X_{n-1}) \\
 &= \prod_{i=1}^n P(X_i|X_{i-1}, \dots, X_1),
 \end{aligned} \tag{1}$$

but for a Bayesian network, based on the edges, we define the full joint distribution as the product of the local conditional distributions:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i|\text{Parents}(X_i)) \tag{2}$$

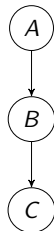
Essence of a Bayesian network: a compact representation of a joint probability distribution in terms of conditional distribution

Probabilistic relationship: standard structures

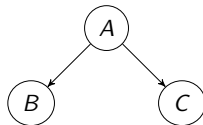
Direct Cause



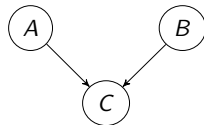
Indirect Cause



Common Cause



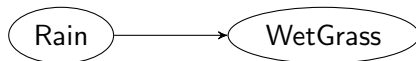
Common effect



Edge: direct cause

Example: We notice the grass in our garden is wet, which might be caused by rain. Construct a Bayesian network to represent their probabilistic relationship.

Solution: The probabilistic graphic model is



Use binary random variables:

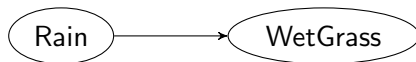
- W where $R_W = \{0, 1\}$ to represent whether the grass is wet (1) or not (0)
- R where $R_r = \{0, 1\}$ to represent whether it has been raining (1) or not (0)

Probabilistic relationship: An edge presents a cause-effect relationship, called direct cause, or conditional dependence between the parent node (cause) and the child node (effect):

$$P(W|R)$$

Probabilistic relationship: direct cause

Solution: The probabilistic graphic model is



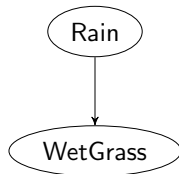
Prior knowledge:

- The probability of raining through a day is
 $P_R(R = 1) = 0.4 \Rightarrow P_R(R = 0) = 1 - P_R(R = 1) = 1 - 0.4 = 0.6$
- The probability that grass gets wet when it rains is 0.9, that is
 $P(W = 1|R = 1) = 0.9 \Rightarrow P(W = 0|R = 1) = 0.1$
- The probability that grass get wet without raining, (e.g., when someone turns on the sprinkler):
 $P(W = 1|R = 0) = 0.2 \Rightarrow P(W = 0|R = 0) = 1 - 0.2 = 0.8$

Probabilistic relationship: direct cause

Solution: The Bayesian network with CPTs:

$P(R = 0)$	$P(R = 1)$
0.6	0.4



R	$P(W = 0)$	$P(W = 1)$
0	0.8	0.2
1	0.1	0.9

Direct cause: Inference

In general the random variables (nodes) fall into two groups:

- **Observed variables:** the ones we have knowledge about
- **Unobserved variables:** ones we do not know about and therefore have to infer the probability.

Question: you observed the grass is wet, what is the probability it rained?

Solution: Based on Bayes theorem, the probability that it rained given the grass is wet can be calculated as

$$P(R = 1|W = 1) = \frac{P(R = 1)P(W = 1|R = 1)}{P(W = 1)}$$

The marginal probability of wet grass can be computed by summing up the joint PMF over the possible values that its parent node can take:

$$\begin{aligned} P(W = 1) &= \sum_{r \in \{0,1\}} P(R = r, W = 1) = \sum_{r \in \{0,1\}} P(W = 1|R = r)P(R = r) \\ &= 0.9 \times 0.4 + 0.2 \times 0.6 = 0.48 \end{aligned}$$

Direct cause: Inference

$$\begin{aligned}P(R = 1|W = 1) &= \frac{P(R = 1)P(W = 1|R = 1)}{P(W = 1)} \\&= \frac{0.9 \times 0.4}{0.48} = 0.75\end{aligned}$$

The above inference is called diagnosis, i.e., to obtain $P(\text{Cause}|\text{Effect})$

Interpretation: Knowing that the grass is wet increased the probability of it rained from 0.4 to 0.75.

Conditional independence

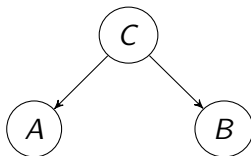
Independence: Two sets of variables A and B are independent iff $P(A) = P(A|B)$ or equivalently $P(A, B) = P(A)P(B)$



Conditional Independence: Two random variables A and B are conditionally independent if they are independent given a third random variable C , written as:

$$(A \perp\!\!\!\perp B) \mid C \iff P(A, B|C) = P(A|C)P(B|C)$$

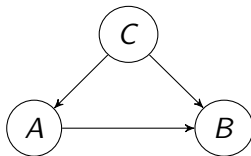
Graphically can be represented as



Why conditional independence matters?

The full joint distribution of three random variables is:

$$P(A, B, C) = P(C)P(A|C)P(B|A, C)$$



Suppose A , B and C can take one of 2 values, $\{0, 1\}$. Modelling the complete joint distribution requires $1 + 2 + 4 = 7$ parameters. In contrast, using the conditional independence as specified in page 17, we need $1 + 2 + 2 = 5$ parameters.

Markov Condition of Bayesian networks

Markov Condition: Each random variable X is conditionally independent of its non-descendants, given its parents.

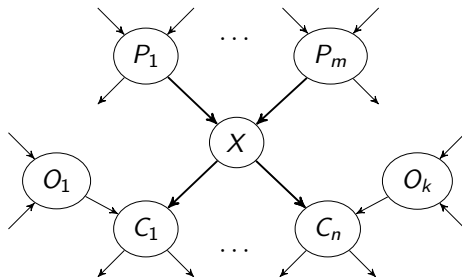
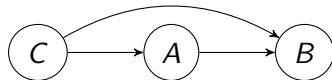
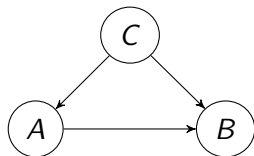
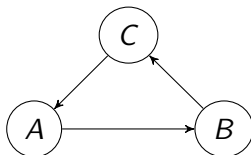


Figure 3: P_1 , P_m are parents of node X , C_1 and C_n are the descendants or children of X . All other nodes, here, denoted as O_1 and O_k are neither parents or children. Therefore, given P_1 and P_m , X is conditionally independent of all other nodes O_1 and O_k . This figure also illustrates an Markov Blanket of X , which is defined as X 's parents + children + children's parents. The Markov blanket is sufficient enough for inferring X .

What do you mean?

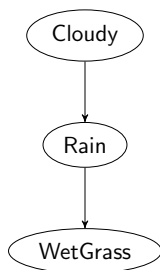
A **standard** Bayesian network only allows the four standard structures in page 11, and the following structures are **NOT** allowed



Take home message: a node is independent of its ancestors given its parents, where the ancestor/parent relationship is with respect to some fixed topological ordering of the nodes.

Probabilistic relationship: Indirect cause

Consider Cloudy or not will influence rain, we have

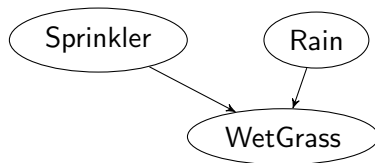


Here since **WetGrass** is independent of **Cloudy** given **Rain**,
According to equation 2, we have the joint distribution:

$$P(C, R, W) = P(C)P(R|C)P(W|R)$$

Probabilistic relationship: Common effect

Suppose we identify another cause of wet grass, the sprinkler, we can then model the relationship as a converging connection (Common effects):



The joint probability distribution:

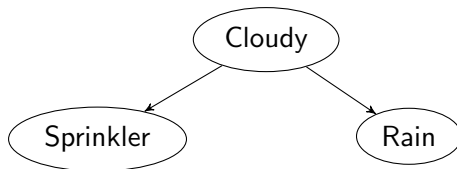
$$P(S, R, W) = P(S)P(R)P(W|S, R)$$

If neither W nor any of its descendants are observed, S and R are independent.

Explain away: If we can confirm one cause (e.g., rain is true) of an observation (e.g., grass is wet), it reduces the need to invoke alternative causes, e.g., sprinkler is on.

Probabilistic relationship: Common cause

It is natural to suppose the cloudy weather influence our decision to turn on the sprinkler or not and cloudy weather will also influence the chance of rain, we model this “common cause” probabilistic relationship using:



The joint probability distribution:

$$P(C, S, R) = P(C)P(S|C)P(R|C)$$

If C is observed, then S and R are independent.

Further reading

- Probabilistic Graphical Models: Chapters 3.1-3.3 (Please find the online version yourself.)
- Artificial Intelligence: Foundations of Computational Agents, Chapter 8